The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Shell-Generating Substrate Lift Dynamics
from Lift-Generating Substrate Precursor Dynamics

Abstract

The new shell-restriction inevitability theorem on the active primitive-reservoir observer-reduction line moved the honest burden one layer lower again. The primitive substrate shell-restriction package is no longer the unexplained stopping point there; inside the shell-generating substrate lift class it is already forced. What therefore remains open is not another support-level or reservoir-level restatement, not a reopening of stationary reduction, stationary Hessians, spectral generators, or selector mechanics, and not any change to the benchmark one-metric package. The remaining honest burden is to derive or justify the shell-generating substrate lift dynamics themselves from still more primitive substrate structure, or else to prove a sharper inevitability theorem at that deeper level.

The present note takes the deeper origin-side route. For each sector it introduces lift-generating substrate precursor dynamics: a compact convex precursor body over the lifted state space, a smooth precursor restriction section with unique precursor minimizers on fixed-lift and fixed-substrate fibers, affine precursor observer-data and shell-response readouts factoring through the lift readouts, symmetry and time reversal, a precursor exact shell projecting diffeomorphically to the exact lifted shell, a precursor shell chart to the same reservoir body, a precursor shell-internal support recorder, a precursor support shell with zero-core / hidden-seed decomposition, a precursor support-shell chart to the same equation variables, a precursor hidden charge, an observer-compatible exact precursor zero-response realization, fixed-data solvability, and coercive control on precursor data-invisible directions. The shell-generating substrate lift dynamics are then recovered canonically by projected exact-shell transport from that deeper precursor package.

Because the induced lift package is exactly the one already used by the shell-restriction inevitability theorem, the downstream shell-restriction and reservoir-facing consequences remain available unchanged. The same benchmark one-metric package remains fixed throughout: the operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. The scope remains disciplined. This note is an origin-side theorem inside the active derivational program of Æther-flow Interpretation of Relativity, not a license for stronger promotion language. The next honest burden is deeper again: derive or justify the precursor dynamics themselves from still more primitive substrate structure, or else prove a sharper inevitability theorem there.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity now has a sharper next task than another package-preserving relay. The immediately preceding theorem already showed that the primitive substrate shell-restriction package is no longer the disciplined place to stop on the active primitive-reservoir observer-reduction line. Once shell-generating substrate lift dynamics

are granted, the shell-restriction package becomes canonical and is forced within that class. That result matters precisely because it moves the honest burden below the shell-restriction layer. The next manuscript therefore cannot honestly be another support-level or reservoir-level reformulation, another reopening of stationary reduction, stationary Hessians, spectral generators, or selector layers, or any change to the one operative metric and the ordinary matter route. The next manuscript has to address the shell-generating lift dynamics themselves.

There are two disciplined ways to do that. One can descend by one more explicit origin layer and derive the shell-generating lift package as the exact projected consequence of still deeper substrate data. Or one can prove that, inside an even sharper deeper class, the shell-generating lift package is itself forced. The present note takes the first route. It introduces a more primitive precursor layer below the current lift dynamics and proves that the entire shell-generating substrate lift package arises there as a projected exact-shell transport. In that sense the lift layer is no longer left as a free insertion on the active line.

That is the correct scope for the next main-line step. The point is not to change the benchmark exact-relativistic theory statement, and it is not to reopen already-settled line-routing decisions. The point is to remove the lift package itself as the next unexplained insertion on the active line. The positive gain is therefore narrower than a completed first-principles substrate derivation and stronger than another same-output restatement of already recorded lower layers.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not alter the benchmark exact-closure package. It does not introduce a second operative metric or a new low-energy matter law. It only strengthens the deeper origin-side explanation of why the shell-generating substrate lift dynamics used by the previous theorem should exist.

2. Fixed Primitive Continuation Data

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *Fixed shell extractors*

$$\mathscr{E}_\sigma^{\mathrm{cur}} : Z_\sigma^{\mathrm{cur}} \to Y_\sigma^{\mathrm{cur}}, \qquad \mathscr{E}_\sigma^{\mathrm{eq}} : Z_\sigma^{\mathrm{eq}} \to Y_\sigma^{\mathrm{eq}},$$

together with the recorded shell composites

$$\mathcal{J}_\sigma^{\mathrm{prim}} = \mathscr{E}_\sigma^{\mathrm{cur}} \circ \mathfrak{C}_\sigma^{\mathrm{cur}}, \qquad \Theta_\sigma^{\mathrm{prim}} = \mathscr{E}_\sigma^{\mathrm{eq}} \circ \mathfrak{C}_\sigma^{\mathrm{eq}}. \tag{1}$$

4. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{2}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Remark 1. *Definition 1 is not reopened here. The primitive continuation data, the one operative metric, and the ordinary matter route remain fixed. The present note only addresses the deeper origin-side status of the shell-generating substrate lift package that now sits immediately above the shell-restriction theorem on the active line.*

3. Recorded Shell-Generating Lift Target

Definition 2 (Shell-generating substrate lift dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* shell-generating substrate lift dynamics package *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean lifted state space*

$$\mathcal{L}_\sigma^{\mathrm{lift}},$$

a compact convex lifted body

$$\mathcal{B}_\sigma^{\mathrm{lift}} \subset \mathcal{L}_\sigma^{\mathrm{lift}}$$

with nonempty interior, an affine surjection

$$\Pi_\sigma^{\mathrm{lift}} : \mathcal{L}_\sigma^{\mathrm{lift}} \to \mathcal{M}_\sigma^{\mathrm{sub}},$$

and the projected substrate body

$$\mathcal{B}_\sigma^{\mathrm{sub}} \equiv \Pi_\sigma^{\mathrm{lift}}(\mathcal{B}_\sigma^{\mathrm{lift}}) \subset \mathcal{M}_\sigma^{\mathrm{sub}},$$

assumed compact convex with nonempty interior. There is a smooth selection

$$\mathfrak{s}_\sigma^{\mathrm{lift}} : \mathcal{B}_\sigma^{\mathrm{sub}} \to \mathcal{B}_\sigma^{\mathrm{lift}}$$

satisfying

$$\Pi_\sigma^{\mathrm{lift}} \circ \mathfrak{s}_\sigma^{\mathrm{lift}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{sub}}}.$$

2. *The same finite-dimensional Euclidean substrate restriction target*

$$E_\sigma^{\mathrm{sub}},$$

the same positive-definite symmetric substrate pairing

$$\mathfrak{s}_\sigma^{\mathrm{sub}} : E_\sigma^{\mathrm{sub}} \times E_\sigma^{\mathrm{sub}} \to \mathbb{R},$$

and a smooth lifted restriction section

$$\Gamma_\sigma^{\mathrm{lift}} : \mathcal{L}_\sigma^{\mathrm{lift}} \to E_\sigma^{\mathrm{sub}}.$$

Define the lifted restriction functional

$$\mathscr{V}_\sigma^{\mathrm{lift}}(\ell) \equiv \mathfrak{s}_\sigma^{\mathrm{sub}}\big(\Gamma_\sigma^{\mathrm{lift}}(\ell), \Gamma_\sigma^{\mathrm{lift}}(\ell)\big). \tag{3}$$

Define the exact lifted shell

$$\mathcal{Z}_\sigma^{\mathrm{lift}} \equiv \big(\Gamma_\sigma^{\mathrm{lift}}\big)^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{lift}}. \tag{4}$$

For each $m \in \mathcal{B}_\sigma^{\mathrm{sub}}$, assume $\mathfrak{s}_\sigma^{\mathrm{lift}}(m)$ is the unique minimizer of $\mathscr{V}_\sigma^{\mathrm{lift}}$ on the fiber

$$\big(\Pi_\sigma^{\mathrm{lift}}\big)^{-1}(m) \cap \mathcal{B}_\sigma^{\mathrm{lift}},$$

and that there exists $\mu_\sigma^{\mathrm{fib}} > 0$ such that

$$\mathscr{V}_\sigma^{\mathrm{lift}}(\ell) \geq \mathscr{V}_\sigma^{\mathrm{lift}}\big(\mathfrak{s}_\sigma^{\mathrm{lift}}(m)\big) + \mu_\sigma^{\mathrm{fib}}\ \mathrm{dist}\big(\ell, \mathfrak{s}_\sigma^{\mathrm{lift}}(m)\big)^2 \tag{5}$$

for every $m \in \mathcal{B}_\sigma^{\mathrm{sub}}$ and every $\ell \in \big(\Pi_\sigma^{\mathrm{lift}}\big)^{-1}(m) \cap \mathcal{B}_\sigma^{\mathrm{lift}}$. Assume there exists $\mu_\sigma^{\mathrm{sub}} > 0$ such that

$$\mathscr{V}_\sigma^{\mathrm{lift}}\big(\mathfrak{s}_\sigma^{\mathrm{lift}}(m)\big) \geq \mu_\sigma^{\mathrm{sub}}\ \mathrm{dist}\big(m, \Pi_\sigma^{\mathrm{lift}}(\mathcal{Z}_\sigma^{\mathrm{lift}})\big)^2 \qquad \textit{for all } m \in \mathcal{B}_\sigma^{\mathrm{sub}}. \tag{6}$$

3. *Affine lifted observer-data and shell-response readouts*

$$\Lambda_\sigma^{\mathrm{lift,dat}} : \mathcal{L}_\sigma^{\mathrm{lift}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}, \qquad \Lambda_\sigma^{\mathrm{lift,sh}} : \mathcal{L}_\sigma^{\mathrm{lift}} \to E_\sigma^{\mathrm{amb}},$$

a finite-dimensional Euclidean support target

$$S_\sigma^{\mathrm{sup,res}},$$

a positive-definite symmetric support pairing

$$\mathfrak{m}_\sigma^{\mathrm{sup,res}} : S_\sigma^{\mathrm{sup,res}} \times S_\sigma^{\mathrm{sup,res}} \to \mathbb{R},$$

a finite-dimensional real hidden target

$$Q_\sigma^{\mathrm{eqn}},$$

and a positive-definite symmetric hidden pairing

$$\mathfrak{b}_\sigma^{\mathrm{eqn}} : Q_\sigma^{\mathrm{eqn}} \times Q_\sigma^{\mathrm{eqn}} \to \mathbb{R}.$$

There is also a compact symmetry group

$$\mathbb{G}_\sigma^{\mathrm{lift}}$$

and an involution

$$\vartheta_\sigma^{\mathrm{lift}},$$

acting orthogonally on $\mathcal{L}_\sigma^{\mathrm{lift}}$, $\mathcal{M}_\sigma^{\mathrm{sub}}$, $\mathcal{Y}_\sigma^{\mathrm{obs}}$, E_σ^{amb}, *and* $S_\sigma^{\mathrm{sup,res}}$, *and by* $\mathfrak{b}_\sigma^{\mathrm{eqn}}$*-isometries on* Q_σ^{eqn}, *preserving* $\mathcal{B}_\sigma^{\mathrm{lift}}$, $\Pi_\sigma^{\mathrm{lift}}$, $\mathfrak{s}_\sigma^{\mathrm{lift}}$, $\Gamma_\sigma^{\mathrm{lift}}$, $\Lambda_\sigma^{\mathrm{lift,dat}}$, *and* $\Lambda_\sigma^{\mathrm{lift,sh}}$.

4. *The restriction of* $\Pi_\sigma^{\mathrm{lift}}$ *to* $\mathcal{Z}_\sigma^{\mathrm{lift}}$ *is a diffeomorphism onto its image*

$$\mathcal{B}_\sigma^{\mathrm{sub}} \supset \mathcal{Z}_\sigma^{\mathrm{sub}} \equiv \Pi_\sigma^{\mathrm{lift}}(\mathcal{Z}_\sigma^{\mathrm{lift}}).$$

There is a Euclidean lifted shell chart

$$\Xi_\sigma^{\mathrm{lift}} : \mathcal{Z}_\sigma^{\mathrm{lift}} \to \mathcal{U}_\sigma^{\mathrm{sup,res}}$$

whose shell-body image

$$\mathcal{B}_\sigma^{\mathrm{sup,res}} \equiv \Xi_\sigma^{\mathrm{lift}}(\mathcal{Z}_\sigma^{\mathrm{lift}})$$

is compact convex with nonempty interior. There exist affine maps

$$\Lambda_\sigma^{\mathrm{sup,dat}} : \mathcal{U}_\sigma^{\mathrm{sup,res}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}, \qquad \Lambda_\sigma^{\mathrm{sup,sh}} : \mathcal{U}_\sigma^{\mathrm{sup,res}} \to E_\sigma^{\mathrm{amb}},$$

and a smooth support section

$$\Theta_\sigma^{\mathrm{sup,res}} : \mathcal{U}_\sigma^{\mathrm{sup,res}} \to S_\sigma^{\mathrm{sup,res}}$$

such that

$$\Lambda_\sigma^{\mathrm{sup,dat}} \circ \Xi_\sigma^{\mathrm{lift}} = \Lambda_\sigma^{\mathrm{lift,dat}}\big|_{\mathcal{Z}_\sigma^{\mathrm{lift}}}, \qquad \Lambda_\sigma^{\mathrm{sup,sh}} \circ \Xi_\sigma^{\mathrm{lift}} = \Lambda_\sigma^{\mathrm{lift,sh}}\big|_{\mathcal{Z}_\sigma^{\mathrm{lift}}}. \tag{7}$$

There is a smooth lifted shell-internal support recorder

$$\Theta_\sigma^{\mathrm{lift,sup}} : \mathcal{Z}_\sigma^{\mathrm{lift}} \to S_\sigma^{\mathrm{sup,res}}$$

satisfying

$$\Theta_\sigma^{\mathrm{sup,res}} \circ \Xi_\sigma^{\mathrm{lift}} = \Theta_\sigma^{\mathrm{lift,sup}}. \tag{8}$$

Define the exact lifted support shell

$$\mathcal{Z}_\sigma^{\mathrm{lift,sup}} \equiv \left(\Theta_\sigma^{\mathrm{lift,sup}}\right)^{-1}(0) \cap \mathcal{Z}_\sigma^{\mathrm{lift}}, \tag{9}$$

and assume

$$\mathcal{Z}_\sigma^{\mathrm{sup,res}} \equiv \big(\Theta_\sigma^{\mathrm{sup,res}}\big)^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{sup,res}} = \Xi_\sigma^{\mathrm{lift}}\big(\mathcal{Z}_\sigma^{\mathrm{lift,sup}}\big). \tag{10}$$

Assume there exists $\nu_\sigma^{\mathrm{sup}} > 0$ *such that*

$$\mathfrak{m}_\sigma^{\mathrm{sup,res}}\big(\Theta_\sigma^{\mathrm{sup,res}}(u), \Theta_\sigma^{\mathrm{sup,res}}(u)\big) \geq \nu_\sigma^{\mathrm{sup}}\ \mathrm{dist}\big(u, \mathcal{Z}_\sigma^{\mathrm{sup,res}}\big)^2 \qquad \textit{for all } u \in \mathcal{B}_\sigma^{\mathrm{sup,res}}. \tag{11}$$

The transported symmetry and time-reversal actions act linearly on $\mathcal{U}_\sigma^{\mathrm{sup,res}}$*, preserving* $\mathcal{B}_\sigma^{\mathrm{sup,res}}$*,* $\mathcal{Z}_\sigma^{\mathrm{sup,res}}$*,* $\Theta_\sigma^{\mathrm{sup,res}}$*,* $\Lambda_\sigma^{\mathrm{sup,dat}}$*, and* $\Lambda_\sigma^{\mathrm{sup,sh}}$*.*

5. *A closed embedded lifted support zero core*

$$\mathcal{Z}_\sigma^{0,\mathrm{lift}} \subset \mathcal{Z}_\sigma^{\mathrm{lift,sup}},$$

a finite-dimensional Euclidean hidden-seed space

$$\mathcal{H}_\sigma^{\mathrm{sup}},$$

and a compact convex hidden-seed body

$$D_\sigma^{\mathrm{sup}} \subset \mathcal{H}_\sigma^{\mathrm{sup}}$$

with

$$0 \in \mathrm{int}\, D_\sigma^{\mathrm{sup}}.$$

There is a Euclidean lifted support-shell chart

$$\Psi_\sigma^{\mathrm{lift,sup}} : \mathcal{Z}_\sigma^{\mathrm{lift,sup}} \to \mathcal{U}_\sigma^{\mathrm{eqn}},$$

an affine chart on the lifted support zero core

$$\Psi_\sigma^{0,\mathrm{lift}} : \mathcal{Z}_\sigma^{0,\mathrm{lift}} \to V_\sigma^{\mathrm{eqn},0},$$

and a linear isometric embedding

$$\Psi_\sigma^{\mathrm{hid}} : \mathcal{H}_\sigma^{\mathrm{sup}} \to H_\sigma^{\mathrm{eqn}}$$

such that

$$\mathcal{U}_\sigma^{\mathrm{eqn}} = V_\sigma^{\mathrm{eqn},0} \oplus H_\sigma^{\mathrm{eqn}}$$

is an orthogonal direct sum, the set

$$\mathcal{B}_\sigma^{0,\mathrm{sup}} \equiv \Psi_\sigma^{0,\mathrm{lift}}\big(\mathcal{Z}_\sigma^{0,\mathrm{lift}}\big)$$

is compact convex with nonempty interior in $V_\sigma^{\mathrm{eqn},0}$*, the shell image*

$$\mathcal{B}_\sigma^{\mathrm{eqn}} \equiv \Psi_\sigma^{\mathrm{lift,sup}}\big(\mathcal{Z}_\sigma^{\mathrm{lift,sup}}\big)$$

has nonempty interior in $\mathcal{U}_\sigma^{\mathrm{eqn}}$*, and there is a diffeomorphism*

$$\Phi_\sigma^{\mathrm{lift}} : \mathcal{Z}_\sigma^{0,\mathrm{lift}} \times D_\sigma^{\mathrm{sup}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{lift,sup}} \tag{12}$$

satisfying

$$\Psi_\sigma^{\mathrm{lift,sup}}\big(\Phi_\sigma^{\mathrm{lift}}(z,\eta)\big) = \Psi_\sigma^{0,\mathrm{lift}}(z) + \Psi_\sigma^{\mathrm{hid}}(\eta). \tag{13}$$

The lifted shell readouts become affine in the lifted support-shell chart: there exist affine maps

$$\Lambda_\sigma^{\mathrm{eqn,dat}} : \mathcal{U}_\sigma^{\mathrm{eqn}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}, \qquad \Lambda_\sigma^{\mathrm{eqn,sh}} : \mathcal{U}_\sigma^{\mathrm{eqn}} \to E_\sigma^{\mathrm{amb}},$$

such that

$$\Lambda_\sigma^{\mathrm{eqn,dat}} \circ \Psi_\sigma^{\mathrm{lift,sup}} = \Lambda_\sigma^{\mathrm{lift,dat}}\big|_{\mathcal{Z}_\sigma^{\mathrm{lift,sup}}}, \qquad \Lambda_\sigma^{\mathrm{eqn,sh}} \circ \Psi_\sigma^{\mathrm{lift,sup}} = \Lambda_\sigma^{\mathrm{lift,sh}}\big|_{\mathcal{Z}_\sigma^{\mathrm{lift,sup}}}. \tag{14}$$

The transported symmetry and time-reversal actions act linearly on $\mathcal{U}_\sigma^{\mathrm{eqn}}$*, preserving* $V_\sigma^{\mathrm{eqn},0}$*,* H_σ^{eqn}*, and* $\mathcal{B}_\sigma^{\mathrm{eqn}}$*.*

6. *A linear hidden charge map*

$$\mathcal{Q}_\sigma^{\mathrm{lift}} : \mathcal{H}_\sigma^{\mathrm{sup}} \to Q_\sigma^{\mathrm{eqn}}$$

such that there exists $\lambda_\sigma^{\mathrm{eqn}} > 0$ *with*

$$\mathfrak{b}_\sigma^{\mathrm{eqn}}\big(\mathcal{Q}_\sigma^{\mathrm{lift}}\eta, \mathcal{Q}_\sigma^{\mathrm{lift}}\eta\big) \geq \lambda_\sigma^{\mathrm{eqn}} \|\eta\|^2 \qquad \textit{for all } \eta \in \mathcal{H}_\sigma^{\mathrm{sup}}. \tag{15}$$

Define the lifted support-shell hidden recorder by

$$\mathcal{R}_\sigma^{\mathrm{lift}}\big(\Phi_\sigma^{\mathrm{lift}}(z,\eta)\big) \equiv \mathcal{Q}_\sigma^{\mathrm{lift}}(\eta). \tag{16}$$

7. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth observer-compatible exact lifted zero-response realization

$$\zeta_\sigma^{\mathrm{lift}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma^{0,\mathrm{lift}}$$

whose image is exactly

$$\mathcal{Z}_\sigma^{0,\mathrm{lift}} \cap \big(\Lambda_\sigma^{\mathrm{lift,sh}}\big)^{-1}(0),$$

and such that

$$\Lambda_\sigma^{\mathrm{lift,dat}}\big(\zeta_\sigma^{\mathrm{lift}}(w)\big) = \big(\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)\big) \tag{17}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$*.*

8. *Fixed-data solvability on the lifted support zero core:*

$$\Lambda_\sigma^{\mathrm{lift,dat}}\big(\mathcal{Z}_\sigma^{0,\mathrm{lift}}\big) = \Lambda_\sigma^{\mathrm{lift,dat}}\Big(\mathcal{Z}_\sigma^{0,\mathrm{lift}} \cap \big(\Lambda_\sigma^{\mathrm{lift,sh}}\big)^{-1}(0)\Big). \tag{18}$$

9. *The inert lifted support-zero-core kernel*

$$K_\sigma^{0,\mathrm{lift}} \equiv \ker\big(D\Lambda_\sigma^{\mathrm{lift,dat}}|_{\mathcal{Z}_\sigma^{0,\mathrm{lift}}}\big) \cap \ker\big(D\Lambda_\sigma^{\mathrm{lift,sh}}|_{\mathcal{Z}_\sigma^{0,\mathrm{lift}}}\big)$$

and the orthogonal projector

$$P_\sigma^{\mathrm{lift,co}} : V_\sigma^{\mathrm{eqn},0} \to (\widetilde{K}_\sigma^{0,\mathrm{lift}})^\perp \cap V_\sigma^{\mathrm{eqn},0}, \qquad \widetilde{K}_\sigma^{0,\mathrm{lift}} \equiv D\Psi_\sigma^{0,\mathrm{lift}}(K_\sigma^{0,\mathrm{lift}}).$$

There exists $\kappa_\sigma^{\mathrm{eqn}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta\ell \in T_{\zeta_\sigma^{\mathrm{lift}}(w)}\mathcal{Z}_\sigma^{\mathrm{lift,sup}}$$

satisfying

$$D\Lambda_\sigma^{\mathrm{lift,dat}}\,\delta\ell = 0,$$

if

$$D\Psi_\sigma^{\mathrm{lift,sup}}(\delta\ell) = \delta z + \delta\eta, \qquad \delta z \in V_\sigma^{\mathrm{eqn},0}, \quad \delta\eta \in H_\sigma^{\mathrm{eqn}},$$

then

$$\|D\Lambda_\sigma^{\mathrm{lift,sh}}\,\delta\ell\|^2 + \mathfrak{b}_\sigma^{\mathrm{eqn}}\big(\mathcal{Q}_\sigma^{\mathrm{lift}}((\Psi_\sigma^{\mathrm{hid}})^{-1}\delta\eta), \mathcal{Q}_\sigma^{\mathrm{lift}}((\Psi_\sigma^{\mathrm{hid}})^{-1}\delta\eta)\big) \geq \kappa_\sigma^{\mathrm{eqn}}\left(\|P_\sigma^{\mathrm{lift,co}}(\delta z)\|^2 + \|\delta\eta\|^2\right). \tag{19}$$

Remark 2. *Definition 2 is the precise package that now has to be explained one layer deeper. The benchmark claim boundary says that such an explanation may sharpen the origin-side derivational line, but it may not change the one operative metric, enlarge the low-energy matter law, or blur the distinction between exact relativistic adoption and first-principles derivation.*

4. Lift-Generating Substrate Precursor Dynamics

Definition 3 (Lift-generating substrate precursor dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* lift-generating substrate precursor dynamics package *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean precursor state space*

$$\mathcal{P}_\sigma^{\mathrm{prec}},$$

a compact convex precursor body

$$\mathcal{B}_\sigma^{\mathrm{prec}} \subset \mathcal{P}_\sigma^{\mathrm{prec}}$$

with nonempty interior, an affine surjection

$$\Pi_\sigma^{\mathrm{prec}} : \mathcal{P}_\sigma^{\mathrm{prec}} \to \mathcal{L}_\sigma^{\mathrm{lift}},$$

and the projected lift body

$$\mathcal{B}_\sigma^{\mathrm{lift}} \equiv \Pi_\sigma^{\mathrm{prec}}(\mathcal{B}_\sigma^{\mathrm{prec}}) \subset \mathcal{L}_\sigma^{\mathrm{lift}},$$

assumed compact convex with nonempty interior. There is an affine surjection

$$\Pi_\sigma^{\mathrm{lift}} : \mathcal{L}_\sigma^{\mathrm{lift}} \to \mathcal{M}_\sigma^{\mathrm{sub}},$$

and the projected substrate body

$$\mathcal{B}_\sigma^{\mathrm{sub}} \equiv \Pi_\sigma^{\mathrm{lift}}(\mathcal{B}_\sigma^{\mathrm{lift}}) \subset \mathcal{M}_\sigma^{\mathrm{sub}},$$

also assumed compact convex with nonempty interior.

There is a smooth precursor lift-fiber minimizer

$$\mathfrak{f}_\sigma^{\mathrm{prec}} : \mathcal{B}_\sigma^{\mathrm{lift}} \to \mathcal{B}_\sigma^{\mathrm{prec}}$$

satisfying

$$\Pi_\sigma^{\mathrm{prec}} \circ \mathfrak{f}_\sigma^{\mathrm{prec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{lift}}},$$

and a smooth precursor substrate-fiber minimizer

$$\mathfrak{m}_\sigma^{\mathrm{prec}} : \mathcal{B}_\sigma^{\mathrm{sub}} \to \mathcal{B}_\sigma^{\mathrm{prec}}$$

satisfying

$$\Pi_\sigma^{\mathrm{lift}} \circ \Pi_\sigma^{\mathrm{prec}} \circ \mathfrak{m}_\sigma^{\mathrm{prec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{sub}}}.$$

Define

$$\mathfrak{s}_\sigma^{\mathrm{lift}} \equiv \Pi_\sigma^{\mathrm{prec}} \circ \mathfrak{m}_\sigma^{\mathrm{prec}}. \tag{20}$$

Assume

$$\mathfrak{f}_\sigma^{\mathrm{prec}}\big(\mathfrak{s}_\sigma^{\mathrm{lift}}(m)\big) = \mathfrak{m}_\sigma^{\mathrm{prec}}(m) \qquad \textit{for every } m \in \mathcal{B}_\sigma^{\mathrm{sub}}. \tag{21}$$

2. *The same finite-dimensional Euclidean substrate restriction target*

$$E_\sigma^{\mathrm{sub}},$$

the same positive-definite symmetric substrate pairing

$$\mathfrak{s}_\sigma^{\mathrm{sub}} : E_\sigma^{\mathrm{sub}} \times E_\sigma^{\mathrm{sub}} \to \mathbb{R},$$

and a smooth precursor restriction section

$$\Gamma_\sigma^{\mathrm{prec}} : \mathcal{P}_\sigma^{\mathrm{prec}} \to E_\sigma^{\mathrm{sub}}.$$

Define the precursor restriction functional

$$\mathscr{W}_\sigma^{\mathrm{prec}}(p) \equiv \mathfrak{s}_\sigma^{\mathrm{sub}}\big(\Gamma_\sigma^{\mathrm{prec}}(p), \Gamma_\sigma^{\mathrm{prec}}(p)\big), \tag{22}$$

together with the exact precursor shell

$$\mathcal{Z}_\sigma^{\mathrm{prec}} \equiv \big(\Gamma_\sigma^{\mathrm{prec}}\big)^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{prec}}. \tag{23}$$

Define the induced lifted restriction section

$$\Gamma_\sigma^{\mathrm{lift}} \equiv \Gamma_\sigma^{\mathrm{prec}} \circ \mathfrak{f}_\sigma^{\mathrm{prec}}, \tag{24}$$

and the induced lifted restriction functional

$$\mathscr{V}_\sigma^{\mathrm{lift}}(\ell) \equiv \mathfrak{s}_\sigma^{\mathrm{sub}}\big(\Gamma_\sigma^{\mathrm{lift}}(\ell), \Gamma_\sigma^{\mathrm{lift}}(\ell)\big) = \mathscr{W}_\sigma^{\mathrm{prec}}\big(\mathfrak{f}_\sigma^{\mathrm{prec}}(\ell)\big). \tag{25}$$

Assume that, for each $\ell \in \mathcal{B}_\sigma^{\mathrm{lift}}$, $\mathfrak{f}_\sigma^{\mathrm{prec}}(\ell)$ *is the unique minimizer of* $\mathscr{W}_\sigma^{\mathrm{prec}}$ *on the fiber*

$$\big(\Pi_\sigma^{\mathrm{prec}}\big)^{-1}(\ell) \cap \mathcal{B}_\sigma^{\mathrm{prec}},$$

and, for each $m \in \mathcal{B}_\sigma^{\mathrm{sub}}$, $\mathfrak{m}_\sigma^{\mathrm{prec}}(m)$ *is the unique minimizer of* $\mathscr{W}_\sigma^{\mathrm{prec}}$ *on the composite fiber*

$$\big(\Pi_\sigma^{\mathrm{lift}} \circ \Pi_\sigma^{\mathrm{prec}}\big)^{-1}(m) \cap \mathcal{B}_\sigma^{\mathrm{prec}}.$$

Assume that $\mathfrak{s}_\sigma^{\mathrm{lift}}(m)$ *is therefore the unique minimizer of* $\mathscr{V}_\sigma^{\mathrm{lift}}$ *on*

$$\big(\Pi_\sigma^{\mathrm{lift}}\big)^{-1}(m) \cap \mathcal{B}_\sigma^{\mathrm{lift}},$$

and that there exists $\mu_\sigma^{\mathrm{fib}} > 0$ *such that*

$$\mathscr{V}_\sigma^{\mathrm{lift}}(\ell) \geq \mathscr{V}_\sigma^{\mathrm{lift}}\big(\mathfrak{s}_\sigma^{\mathrm{lift}}(m)\big) + \mu_\sigma^{\mathrm{fib}}\ \mathrm{dist}\big(\ell, \mathfrak{s}_\sigma^{\mathrm{lift}}(m)\big)^2 \tag{26}$$

for every $m \in \mathcal{B}_\sigma^{\mathrm{sub}}$ *and every* $\ell \in \big(\Pi_\sigma^{\mathrm{lift}}\big)^{-1}(m) \cap \mathcal{B}_\sigma^{\mathrm{lift}}$. *Assume there exists* $\mu_\sigma^{\mathrm{sub}} > 0$ *such that*

$$\mathscr{V}_\sigma^{\mathrm{lift}}\big(\mathfrak{s}_\sigma^{\mathrm{lift}}(m)\big) \geq \mu_\sigma^{\mathrm{sub}}\ \mathrm{dist}\big(m, \Pi_\sigma^{\mathrm{lift}}(\Pi_\sigma^{\mathrm{prec}}(\mathcal{Z}_\sigma^{\mathrm{prec}}))\big)^2 \qquad \textit{for all } m \in \mathcal{B}_\sigma^{\mathrm{sub}}. \tag{27}$$

3. *Affine lifted observer-data and shell-response readouts*

$$\Lambda_\sigma^{\mathrm{lift,dat}} : \mathcal{L}_\sigma^{\mathrm{lift}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}, \qquad \Lambda_\sigma^{\mathrm{lift,sh}} : \mathcal{L}_\sigma^{\mathrm{lift}} \to E_\sigma^{\mathrm{amb}},$$

and affine precursor observer-data and shell-response readouts

$$\Lambda_\sigma^{\mathrm{prec,dat}} : \mathcal{P}_\sigma^{\mathrm{prec}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}, \qquad \Lambda_\sigma^{\mathrm{prec,sh}} : \mathcal{P}_\sigma^{\mathrm{prec}} \to E_\sigma^{\mathrm{amb}},$$

satisfying

$$\Lambda_\sigma^{\mathrm{prec,dat}} = \Lambda_\sigma^{\mathrm{lift,dat}} \circ \Pi_\sigma^{\mathrm{prec}}, \qquad \Lambda_\sigma^{\mathrm{prec,sh}} = \Lambda_\sigma^{\mathrm{lift,sh}} \circ \Pi_\sigma^{\mathrm{prec}}. \tag{28}$$

There is also a finite-dimensional Euclidean support target

$$S_\sigma^{\mathrm{sup,res}},$$

a positive-definite symmetric support pairing

$$\mathfrak{m}_\sigma^{\mathrm{sup,res}} : S_\sigma^{\mathrm{sup,res}} \times S_\sigma^{\mathrm{sup,res}} \to \mathbb{R},$$

a finite-dimensional real hidden target

$$Q_\sigma^{\mathrm{eqn}},$$

and a positive-definite symmetric hidden pairing

$$\mathfrak{b}_\sigma^{\mathrm{eqn}} : Q_\sigma^{\mathrm{eqn}} \times Q_\sigma^{\mathrm{eqn}} \to \mathbb{R}.$$

There is a compact symmetry group

$$\mathbb{G}_\sigma^{\mathrm{prec}}$$

and an involution

$$\vartheta_\sigma^{\mathrm{prec}},$$

acting orthogonally on $\mathcal{P}_\sigma^{\mathrm{prec}}$, $\mathcal{L}_\sigma^{\mathrm{lift}}$, $\mathcal{M}_\sigma^{\mathrm{sub}}$, $\mathcal{Y}_\sigma^{\mathrm{obs}}$, E_σ^{amb}, *and* $S_\sigma^{\mathrm{sup,res}}$, *and by* $\mathfrak{b}_\sigma^{\mathrm{eqn}}$*-isometries on* Q_σ^{eqn}, *preserving* $\mathcal{B}_\sigma^{\mathrm{prec}}$, $\mathcal{B}_\sigma^{\mathrm{lift}}$, $\Pi_\sigma^{\mathrm{prec}}$, $\Pi_\sigma^{\mathrm{lift}}$, $\mathfrak{f}_\sigma^{\mathrm{prec}}$, $\mathfrak{m}_\sigma^{\mathrm{prec}}$, $\Gamma_\sigma^{\mathrm{prec}}$, $\Lambda_\sigma^{\mathrm{lift,dat}}$, $\Lambda_\sigma^{\mathrm{lift,sh}}$, $\Lambda_\sigma^{\mathrm{prec,dat}}$, *and* $\Lambda_\sigma^{\mathrm{prec,sh}}$.

4. *The restriction of* $\Pi_\sigma^{\mathrm{prec}}$ *to* $\mathcal{Z}_\sigma^{\mathrm{prec}}$ *is a diffeomorphism onto its image*

$$\mathcal{Z}_\sigma^{\mathrm{lift}} \equiv \Pi_\sigma^{\mathrm{prec}}(\mathcal{Z}_\sigma^{\mathrm{prec}}) \subset \mathcal{B}_\sigma^{\mathrm{lift}}.$$

There is a Euclidean precursor shell chart

$$\Xi_\sigma^{\mathrm{prec}} : \mathcal{Z}_\sigma^{\mathrm{prec}} \to \mathcal{U}_\sigma^{\mathrm{sup,res}}$$

whose shell-body image

$$\mathcal{B}_\sigma^{\mathrm{sup,res}} \equiv \Xi_\sigma^{\mathrm{prec}}(\mathcal{Z}_\sigma^{\mathrm{prec}})$$

is compact convex with nonempty interior. There exist affine maps

$$\Lambda_\sigma^{\mathrm{sup,dat}} : \mathcal{U}_\sigma^{\mathrm{sup,res}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}, \qquad \Lambda_\sigma^{\mathrm{sup,sh}} : \mathcal{U}_\sigma^{\mathrm{sup,res}} \to E_\sigma^{\mathrm{amb}},$$

and a smooth support section

$$\Theta_\sigma^{\mathrm{sup,res}} : \mathcal{U}_\sigma^{\mathrm{sup,res}} \to S_\sigma^{\mathrm{sup,res}}$$

such that

$$\Lambda_\sigma^{\mathrm{sup,dat}} \circ \Xi_\sigma^{\mathrm{prec}} = \Lambda_\sigma^{\mathrm{prec,dat}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec}}}, \qquad \Lambda_\sigma^{\mathrm{sup,sh}} \circ \Xi_\sigma^{\mathrm{prec}} = \Lambda_\sigma^{\mathrm{prec,sh}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec}}}. \tag{29}$$

There is a smooth precursor shell-internal support recorder

$$\Theta_\sigma^{\mathrm{prec,sup}} : \mathcal{Z}_\sigma^{\mathrm{prec}} \to S_\sigma^{\mathrm{sup,res}}$$

satisfying

$$\Theta_\sigma^{\mathrm{sup,res}} \circ \Xi_\sigma^{\mathrm{prec}} = \Theta_\sigma^{\mathrm{prec,sup}}. \tag{30}$$

Define the exact precursor support shell

$$\mathcal{Z}_\sigma^{\mathrm{prec,sup}} \equiv \big(\Theta_\sigma^{\mathrm{prec,sup}}\big)^{-1}(0) \cap \mathcal{Z}_\sigma^{\mathrm{prec}}, \tag{31}$$

and assume

$$\mathcal{Z}_\sigma^{\mathrm{sup,res}} \equiv \big(\Theta_\sigma^{\mathrm{sup,res}}\big)^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{sup,res}} = \Xi_\sigma^{\mathrm{prec}}(\mathcal{Z}_\sigma^{\mathrm{prec,sup}}). \tag{32}$$

Assume there exists $\nu_\sigma^{\mathrm{sup}} > 0$ *such that*

$$\mathfrak{m}_\sigma^{\mathrm{sup,res}}\big(\Theta_\sigma^{\mathrm{sup,res}}(u), \Theta_\sigma^{\mathrm{sup,res}}(u)\big) \geq \nu_\sigma^{\mathrm{sup}}\ \mathrm{dist}\big(u, \mathcal{Z}_\sigma^{\mathrm{sup,res}}\big)^2 \qquad \textit{for all } u \in \mathcal{B}_\sigma^{\mathrm{sup,res}}. \tag{33}$$

Assume the restriction of $\Pi_\sigma^{\mathrm{prec}}$ *to* $\mathcal{Z}_\sigma^{\mathrm{prec,sup}}$ *is a diffeomorphism onto its image*

$$\mathcal{Z}_\sigma^{\mathrm{lift,sup}} \equiv \Pi_\sigma^{\mathrm{prec}}(\mathcal{Z}_\sigma^{\mathrm{prec,sup}}).$$

The transported symmetry and time-reversal actions act linearly on $\mathcal{U}_\sigma^{\mathrm{sup,res}}$*, preserving* $\mathcal{B}_\sigma^{\mathrm{sup,res}}$*,* $\mathcal{Z}_\sigma^{\mathrm{sup,res}}$*,* $\Theta_\sigma^{\mathrm{sup,res}}$*,* $\Lambda_\sigma^{\mathrm{sup,dat}}$*, and* $\Lambda_\sigma^{\mathrm{sup,sh}}$*.*

5. *A closed embedded precursor support zero core*

$$\mathcal{Z}_\sigma^{0,\mathrm{prec}} \subset \mathcal{Z}_\sigma^{\mathrm{prec,sup}},$$

a finite-dimensional Euclidean hidden-seed space

$$\mathcal{H}_\sigma^{\mathrm{sup}},$$

and a compact convex hidden-seed body

$$D_\sigma^{\mathrm{sup}} \subset \mathcal{H}_\sigma^{\mathrm{sup}}$$

with

$$0 \in \operatorname{int} D_\sigma^{\mathrm{sup}}.$$

There is a Euclidean precursor support-shell chart

$$\Psi_\sigma^{\mathrm{prec,sup}} : \mathcal{Z}_\sigma^{\mathrm{prec,sup}} \to \mathcal{U}_\sigma^{\mathrm{eqn}},$$

an affine chart on the precursor support zero core

$$\Psi_\sigma^{0,\mathrm{prec}} : \mathcal{Z}_\sigma^{0,\mathrm{prec}} \to V_\sigma^{\mathrm{eqn},0},$$

and a linear isometric embedding

$$\Psi_\sigma^{\mathrm{hid}} : \mathcal{H}_\sigma^{\mathrm{sup}} \to H_\sigma^{\mathrm{eqn}}$$

such that

$$\mathcal{U}_\sigma^{\mathrm{eqn}} = V_\sigma^{\mathrm{eqn},0} \oplus H_\sigma^{\mathrm{eqn}}$$

is an orthogonal direct sum, the set

$$\mathcal{B}_\sigma^{0,\mathrm{sup}} \equiv \Psi_\sigma^{0,\mathrm{prec}}(\mathcal{Z}_\sigma^{0,\mathrm{prec}})$$

is compact convex with nonempty interior in $V_\sigma^{\mathrm{eqn},0}$*, the shell image*

$$\mathcal{B}_\sigma^{\mathrm{eqn}} \equiv \Psi_\sigma^{\mathrm{prec,sup}}(\mathcal{Z}_\sigma^{\mathrm{prec,sup}})$$

has nonempty interior in $\mathcal{U}_\sigma^{\mathrm{eqn}}$*, and there is a diffeomorphism*

$$\Phi_\sigma^{\mathrm{prec}} : \mathcal{Z}_\sigma^{0,\mathrm{prec}} \times D_\sigma^{\mathrm{sup}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{prec,sup}} \tag{34}$$

satisfying

$$\Psi_\sigma^{\mathrm{prec,sup}}\big(\Phi_\sigma^{\mathrm{prec}}(z,\eta)\big) = \Psi_\sigma^{0,\mathrm{prec}}(z) + \Psi_\sigma^{\mathrm{hid}}(\eta). \tag{35}$$

The precursor shell readouts become affine in the precursor support-shell chart: there exist affine maps

$$\Lambda_\sigma^{\mathrm{eqn,dat}} : \mathcal{U}_\sigma^{\mathrm{eqn}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}, \qquad \Lambda_\sigma^{\mathrm{eqn,sh}} : \mathcal{U}_\sigma^{\mathrm{eqn}} \to E_\sigma^{\mathrm{amb}},$$

such that

$$\Lambda_\sigma^{\mathrm{eqn,dat}} \circ \Psi_\sigma^{\mathrm{prec,sup}} = \Lambda_\sigma^{\mathrm{prec,dat}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec,sup}}}, \qquad \Lambda_\sigma^{\mathrm{eqn,sh}} \circ \Psi_\sigma^{\mathrm{prec,sup}} = \Lambda_\sigma^{\mathrm{prec,sh}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec,sup}}}. \tag{36}$$

Assume the restriction of $\Pi_\sigma^{\mathrm{prec}}$ *to* $\mathcal{Z}_\sigma^{0,\mathrm{prec}}$ *is a diffeomorphism onto its image*

$$\mathcal{Z}_\sigma^{0,\mathrm{lift}} \equiv \Pi_\sigma^{\mathrm{prec}}\big(\mathcal{Z}_\sigma^{0,\mathrm{prec}}\big).$$

The transported symmetry and time-reversal actions act linearly on $\mathcal{U}_\sigma^{\mathrm{eqn}}$*, preserving* $V_\sigma^{\mathrm{eqn},0}$*,* H_σ^{eqn}*, and* $\mathcal{B}_\sigma^{\mathrm{eqn}}$*.*

6. *A linear hidden charge map*

$$\mathcal{Q}_\sigma^{\mathrm{prec}} : \mathcal{H}_\sigma^{\mathrm{sup}} \to Q_\sigma^{\mathrm{eqn}}$$

such that there exists $\lambda_\sigma^{\mathrm{eqn}} > 0$ *with*

$$\mathfrak{b}_\sigma^{\mathrm{eqn}}\big(\mathcal{Q}_\sigma^{\mathrm{prec}}\eta, \mathcal{Q}_\sigma^{\mathrm{prec}}\eta\big) \geq \lambda_\sigma^{\mathrm{eqn}} \|\eta\|^2 \qquad \textit{for all } \eta \in \mathcal{H}_\sigma^{\mathrm{sup}}. \tag{37}$$

Define the precursor support-shell hidden recorder by

$$\mathcal{R}_\sigma^{\mathrm{prec}}\big(\Phi_\sigma^{\mathrm{prec}}(z,\eta)\big) \equiv \mathcal{Q}_\sigma^{\mathrm{prec}}(\eta). \tag{38}$$

7. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth observer-compatible exact precursor zero-response realization

$$\zeta_\sigma^{\mathrm{prec}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma^{0,\mathrm{prec}}$$

whose image is exactly

$$\mathcal{Z}_\sigma^{0,\mathrm{prec}} \cap \big(\Lambda_\sigma^{\mathrm{prec,sh}}\big)^{-1}(0),$$

and such that

$$\Lambda_\sigma^{\mathrm{prec,dat}}\big(\zeta_\sigma^{\mathrm{prec}}(w)\big) = \big(\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)\big) \tag{39}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$*.*

8. *Fixed-data solvability on the precursor support zero core:*

$$\Lambda_\sigma^{\mathrm{prec,dat}}\big(\mathcal{Z}_\sigma^{0,\mathrm{prec}}\big) = \Lambda_\sigma^{\mathrm{prec,dat}}\Big(\mathcal{Z}_\sigma^{0,\mathrm{prec}} \cap \big(\Lambda_\sigma^{\mathrm{prec,sh}}\big)^{-1}(0)\Big). \tag{40}$$

9. *The inert precursor support-zero-core kernel*

$$K_\sigma^{0,\mathrm{prec}} \equiv \ker\big(D\Lambda_\sigma^{\mathrm{prec,dat}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{prec}}}\big) \cap \ker\big(D\Lambda_\sigma^{\mathrm{prec,sh}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{prec}}}\big)$$

and the orthogonal projector

$$P_\sigma^{\mathrm{prec,co}} : V_\sigma^{\mathrm{eqn},0} \to \big(\widetilde{K}_\sigma^{0,\mathrm{prec}}\big)^\perp \cap V_\sigma^{\mathrm{eqn},0}, \qquad \widetilde{K}_\sigma^{0,\mathrm{prec}} \equiv D\Psi_\sigma^{0,\mathrm{prec}}\big(K_\sigma^{0,\mathrm{prec}}\big).$$

There exists $\kappa_\sigma^{\mathrm{eqn}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta p \in T_{\zeta_\sigma^{\mathrm{prec}}(w)} \mathcal{Z}_\sigma^{\mathrm{prec,sup}}$$

satisfying

$$D\Lambda_\sigma^{\mathrm{prec,dat}}\, \delta p = 0,$$

if

$$D\Psi_\sigma^{\mathrm{prec,sup}}(\delta p) = \delta z + \delta\eta, \qquad \delta z \in V_\sigma^{\mathrm{eqn},0}, \quad \delta\eta \in H_\sigma^{\mathrm{eqn}},$$

then

$$\begin{aligned} \|D\Lambda_\sigma^{\mathrm{prec,sh}}\, \delta p\|^2 + \mathfrak{b}_\sigma^{\mathrm{eqn}}\big(\mathcal{Q}_\sigma^{\mathrm{prec}}\big((\Psi_\sigma^{\mathrm{hid}})^{-1}\delta\eta\big), \\ \mathcal{Q}_\sigma^{\mathrm{prec}}\big((\Psi_\sigma^{\mathrm{hid}})^{-1}\delta\eta\big)\big) \\ \geq \kappa_\sigma^{\mathrm{eqn}} \left(\|P_\sigma^{\mathrm{prec,co}}(\delta z)\|^2 + \|\delta\eta\|^2 \right). \end{aligned} \tag{41}$$

Remark 3. *Definition 3 is deliberately deeper than the shell-generating lift package but narrower than a claim of completed substrate microphysics. The lift layer is not discarded. Instead it is shown to arise as the projected exact-shell image of a still more primitive precursor layer whose exact shell, support shell, zero-core / hidden-seed split, compatibility realization, solvability statement, and coercive control all first live at the precursor level.*

5. Canonical Shell-Generating Lift Package

Proposition 1 (The shell-generating substrate lift package is produced canonically)**.** *Assume Definition 3. Define, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*,*

$$\Gamma_\sigma^{\mathrm{lift}} \equiv \Gamma_\sigma^{\mathrm{prec}} \circ \mathfrak{f}_\sigma^{\mathrm{prec}}, \tag{42}$$

the lifted shell chart

$$\Xi_\sigma^{\mathrm{lift}} \equiv \Xi_\sigma^{\mathrm{prec}} \circ \big(\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec}}}\big)^{-1}, \tag{43}$$

the lifted shell-internal support recorder

$$\Theta_\sigma^{\mathrm{lift,sup}} \equiv \Theta_\sigma^{\mathrm{prec,sup}} \circ \big(\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec}}}\big)^{-1}, \tag{44}$$

the projected support shell and projected support zero core

$$\mathcal{Z}_\sigma^{\mathrm{lift,sup}} \equiv \Pi_\sigma^{\mathrm{prec}}\big(\mathcal{Z}_\sigma^{\mathrm{prec,sup}}\big), \qquad \mathcal{Z}_\sigma^{0,\mathrm{lift}} \equiv \Pi_\sigma^{\mathrm{prec}}\big(\mathcal{Z}_\sigma^{0,\mathrm{prec}}\big), \tag{45}$$

the support-shell and zero-core charts

$$\Psi_\sigma^{\mathrm{lift,sup}} \equiv \Psi_\sigma^{\mathrm{prec,sup}} \circ \big(\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec,sup}}}\big)^{-1}, \qquad \Psi_\sigma^{0,\mathrm{lift}} \equiv \Psi_\sigma^{0,\mathrm{prec}} \circ \big(\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{prec}}}\big)^{-1}, \tag{46}$$

the projected split map

$$\Phi_\sigma^{\mathrm{lift}}(z,\eta) \equiv \Pi_\sigma^{\mathrm{prec}}\Big(\Phi_\sigma^{\mathrm{prec}}\big((\Pi_\sigma^{\mathrm{prec}}|_{\mathcal{Z}_\sigma^{0,\mathrm{prec}}})^{-1}(z),\eta\big)\Big), \tag{47}$$

the hidden charge and hidden recorder

$$\mathcal{Q}_\sigma^{\mathrm{lift}} \equiv \mathcal{Q}_\sigma^{\mathrm{prec}}, \qquad \mathcal{R}_\sigma^{\mathrm{lift}}(\Phi_\sigma^{\mathrm{lift}}(z,\eta)) \equiv \mathcal{Q}_\sigma^{\mathrm{prec}}(\eta), \tag{48}$$

and the exact zero-response realization

$$\zeta_\sigma^{\mathrm{lift}} \equiv \Pi_\sigma^{\mathrm{prec}} \circ \zeta_\sigma^{\mathrm{prec}}. \tag{49}$$

Then the resulting data satisfy exactly the shell-generating substrate lift dynamics package of Definition 2.

Proof. We verify the items of Definition 2 in order.

For item (1), the lifted state space $\mathcal{L}_\sigma^{\mathrm{lift}}$, the compact convex lifted body $\mathcal{B}_\sigma^{\mathrm{lift}}$, the affine surjection $\Pi_\sigma^{\mathrm{lift}}$, and the projected substrate body $\mathcal{B}_\sigma^{\mathrm{sub}}$ are already part of Definition 3(1). Equation (20) gives the smooth selection $\mathfrak{s}_\sigma^{\mathrm{lift}}$, and

$$\Pi_\sigma^{\mathrm{lift}} \circ \mathfrak{s}_\sigma^{\mathrm{lift}} = \Pi_\sigma^{\mathrm{lift}} \circ \Pi_\sigma^{\mathrm{prec}} \circ \mathfrak{m}_\sigma^{\mathrm{prec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{sub}}}$$

by Definition 3(1).

For item (2), equation (42) gives the lifted restriction section, and (25) gives its associated lifted restriction functional. If $\ell \in \Pi_\sigma^{\mathrm{prec}}(\mathcal{Z}_\sigma^{\mathrm{prec}})$, choose $p \in \mathcal{Z}_\sigma^{\mathrm{prec}}$ with $\Pi_\sigma^{\mathrm{prec}}(p) = \ell$. Then $\mathscr{W}_\sigma^{\mathrm{prec}}(p) = 0$, so by the precursor lift-fiber minimality of Definition 3(2),

$$\mathscr{V}_\sigma^{\mathrm{lift}}(\ell) = \mathscr{W}_\sigma^{\mathrm{prec}}(\mathfrak{f}_\sigma^{\mathrm{prec}}(\ell)) = 0.$$

Conversely, if $\mathscr{V}_\sigma^{\mathrm{lift}}(\ell) = 0$, then

$$\mathscr{W}_\sigma^{\mathrm{prec}}(\mathfrak{f}_\sigma^{\mathrm{prec}}(\ell)) = 0,$$

so $\mathfrak{f}_\sigma^{\mathrm{prec}}(\ell) \in \mathcal{Z}_\sigma^{\mathrm{prec}}$ and therefore $\ell \in \Pi_\sigma^{\mathrm{prec}}(\mathcal{Z}_\sigma^{\mathrm{prec}})$. Hence

$$\mathcal{Z}_\sigma^{\mathrm{lift}} = (\Gamma_\sigma^{\mathrm{lift}})^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{lift}} = \Pi_\sigma^{\mathrm{prec}}(\mathcal{Z}_\sigma^{\mathrm{prec}}).$$

The fiber-minimizing property of $\mathfrak{s}_\sigma^{\mathrm{lift}}$ and the estimate (5) are exactly the assumptions (26) of Definition 3(2), while (6) is exactly (27).

For item (3), the affine readouts $\Lambda_\sigma^{\mathrm{lift,dat}}$ and $\Lambda_\sigma^{\mathrm{lift,sh}}$, the support target, support pairing, hidden target, hidden pairing, and the symmetry/time-reversal actions on $\mathcal{L}_\sigma^{\mathrm{lift}}$ are already part of Definition 3(3). Those same actions give the induced $\mathbb{G}_\sigma^{\mathrm{lift}}$ and $\vartheta_\sigma^{\mathrm{lift}}$. Because Definition 3(3) says they preserve $\Pi_\sigma^{\mathrm{prec}}$, $\Pi_\sigma^{\mathrm{lift}}$, $\mathfrak{m}_\sigma^{\mathrm{prec}}$, $\mathfrak{f}_\sigma^{\mathrm{prec}}$, $\Gamma_\sigma^{\mathrm{prec}}$, $\Lambda_\sigma^{\mathrm{lift,dat}}$, and $\Lambda_\sigma^{\mathrm{lift,sh}}$, they preserve $\mathcal{B}_\sigma^{\mathrm{lift}}$, $\mathfrak{s}_\sigma^{\mathrm{lift}}$, $\Gamma_\sigma^{\mathrm{lift}}$, and the full lifted readout package required in Definition 2(3).

For item (4), the restriction $\Pi_\sigma^{\mathrm{prec}}|_{\mathcal{Z}_\sigma^{\mathrm{prec}}}$ is a diffeomorphism onto $\mathcal{Z}_\sigma^{\mathrm{lift}}$ by Definition 3(4), so (43) defines the lifted shell chart. Its image is exactly

$$\mathcal{B}_\sigma^{\mathrm{sup,res}} = \Xi_\sigma^{\mathrm{prec}}(\mathcal{Z}_\sigma^{\mathrm{prec}}).$$

Using (28) and (29),

$$\Lambda_\sigma^{\mathrm{sup,dat}} \circ \Xi_\sigma^{\mathrm{lift}} = \Lambda_\sigma^{\mathrm{prec,dat}} \circ \big(\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec}}}\big)^{-1} = \Lambda_\sigma^{\mathrm{lift,dat}}\big|_{\mathcal{Z}_\sigma^{\mathrm{lift}}},$$

and similarly

$$\Lambda_\sigma^{\mathrm{sup,sh}} \circ \Xi_\sigma^{\mathrm{lift}} = \Lambda_\sigma^{\mathrm{lift,sh}}\big|_{\mathcal{Z}_\sigma^{\mathrm{lift}}}.$$

Equation (44) defines the lifted shell-internal support recorder, and (30) becomes exactly (8). Because the restriction of $\Pi_\sigma^{\mathrm{prec}}$ to $\mathcal{Z}_\sigma^{\mathrm{prec,sup}}$ is a diffeomorphism onto $\mathcal{Z}_\sigma^{\mathrm{lift,sup}}$, one has

$$\mathcal{Z}_\sigma^{\mathrm{sup,res}} = \Xi_\sigma^{\mathrm{prec}}\big(\mathcal{Z}_\sigma^{\mathrm{prec,sup}}\big) = \Xi_\sigma^{\mathrm{lift}}\big(\mathcal{Z}_\sigma^{\mathrm{lift,sup}}\big),$$

which is (10). The normal shell stability (11) is exactly (33). The transported symmetry and time-reversal actions on $\mathcal{U}_\sigma^{\mathrm{sup,res}}$ are the same ones already supplied in Definition 3(4).

For item (5), the projected support zero core is $\mathcal{Z}_\sigma^{0,\mathrm{lift}}$, and the restrictions of $\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec,sup}}}$ and $\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{prec}}}$ are diffeomorphisms onto $\mathcal{Z}_\sigma^{\mathrm{lift,sup}}$ and $\mathcal{Z}_\sigma^{0,\mathrm{lift}}$, respectively. Therefore (46) defines the required charts. The orthogonal splitting

$$\mathcal{U}_\sigma^{\mathrm{eqn}} = V_\sigma^{\mathrm{eqn},0} \oplus H_\sigma^{\mathrm{eqn}},$$

the compact convex bodies $\mathcal{B}_\sigma^{0,\mathrm{sup}}$ and $\mathcal{B}_\sigma^{\mathrm{eqn}}$, and the isometric hidden chart $\Psi_\sigma^{\mathrm{hid}}$ are already part of Definition 3(5). The projected split map (47) is a diffeomorphism because $\Phi_\sigma^{\mathrm{prec}}$ is one and $\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec,sup}}}$ is one. Moreover, if

$$p_z \equiv \big(\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{prec}}}\big)^{-1}(z) \in \mathcal{Z}_\sigma^{0,\mathrm{prec}},$$

then

$$\Psi_\sigma^{\mathrm{lift,sup}}\big(\Phi_\sigma^{\mathrm{lift}}(z,\eta)\big) = \Psi_\sigma^{\mathrm{prec,sup}}\big(\Phi_\sigma^{\mathrm{prec}}(p_z,\eta)\big) = \Psi_\sigma^{0,\mathrm{prec}}(p_z) + \Psi_\sigma^{\mathrm{hid}}(\eta) = \Psi_\sigma^{0,\mathrm{lift}}(z) + \Psi_\sigma^{\mathrm{hid}}(\eta),$$

which is exactly (13). The equation-space readouts $\Lambda_\sigma^{\mathrm{eqn,dat}}$ and $\Lambda_\sigma^{\mathrm{eqn,sh}}$ are precisely the affine maps already supplied in Definition 3(5). Using (28), equation (36) becomes (14).

For item (6), (48) defines the hidden charge and hidden recorder, and the positive hidden gap (15) is exactly (37).

For item (7), (49) defines the exact lifted zero-response realization. Since $\zeta_\sigma^{\mathrm{prec}}$ lands in $\mathcal{Z}_\sigma^{0,\mathrm{prec}} \cap (\Lambda_\sigma^{\mathrm{prec,sh}})^{-1}(0)$, its projected image lands in $\mathcal{Z}_\sigma^{0,\mathrm{lift}} \cap (\Lambda_\sigma^{\mathrm{lift,sh}})^{-1}(0)$ by (28). Because $\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{prec}}}$ is a diffeomorphism onto $\mathcal{Z}_\sigma^{0,\mathrm{lift}}$, the image is exactly that zero-response locus. Equation (17) follows immediately from (39), (28), and (49).

For item (8), fixed-data solvability (18) is the projection of (40) through the zero-core diffeomorphism $\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{prec}}}$ together with the factorization (28).

For item (9), define

$$K_\sigma^{0,\mathrm{lift}} \equiv \ker\big(D\Lambda_\sigma^{\mathrm{lift,dat}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{lift}}}\big) \cap \ker\big(D\Lambda_\sigma^{\mathrm{lift,sh}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{lift}}}\big), \qquad \widetilde{K}_\sigma^{0,\mathrm{lift}} \equiv D\Psi_\sigma^{0,\mathrm{lift}}\big(K_\sigma^{0,\mathrm{lift}}\big),$$

and let $P_\sigma^{\mathrm{lift,co}}$ be the orthogonal projector onto $(\widetilde{K}_\sigma^{0,\mathrm{lift}})^\perp \cap V_\sigma^{\mathrm{eqn},0}$. Take $w \in \mathcal{I}_\sigma^{\mathrm{for}}$ and $\delta\ell \in$

$T_{\zeta_\sigma^{\mathrm{lift}}(w)}\mathcal{Z}_\sigma^{\mathrm{lift,sup}}$ satisfying $D\Lambda_\sigma^{\mathrm{lift,dat}}\,\delta\ell = 0$. Set

$$\delta p \equiv D\big(\Pi_\sigma^{\mathrm{prec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prec,sup}}}\big)^{-1}(\delta\ell) \in T_{\zeta_\sigma^{\mathrm{prec}}(w)}\mathcal{Z}_\sigma^{\mathrm{prec,sup}}.$$

Then $D\Lambda_\sigma^{\mathrm{prec,dat}}\,\delta p = 0$ by (28). Writing

$$D\Psi_\sigma^{\mathrm{lift,sup}}(\delta\ell) = \delta z + \delta\eta, \qquad \delta z \in V_\sigma^{\mathrm{eqn},0}, \quad \delta\eta \in H_\sigma^{\mathrm{eqn}},$$

equation (41) becomes exactly (19). This completes the verification of Definition 2. □

Remark 4. *Proposition 1 gives more than another same-output relay. Every constituent of the shell-generating substrate lift package is now an explicit projected or transported image of deeper precursor data. The lift layer has therefore lost its status as a free insertion on the active line.*

6. Benchmark Preservation

Theorem 1 (The same benchmark one-metric package is preserved)**.** *Assume Definition 1 and, for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, lift-generating substrate precursor dynamics in the sense of Definition 3. Then the benchmark output statements of* (2) *remain unchanged:*

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi]$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ and every matter configuration $\psi \in \Psi$. Consequently the deeper precursor theorem introduces neither a second operative metric nor an enlarged low-energy matter law.

Proof. The present note changes only the deeper origin-side status of the shell-generating substrate lift package. It does not alter the fixed primitive shell $\mathcal{S}_{\mathrm{prim}}$, the recorded metric map $\mathscr{G}_{\mathrm{out}}$, or the recorded matter map $\mathscr{T}_{\mathrm{out}}$. Those remain the continuation data of Definition 1, so (2) continues to hold exactly. □

7. Main Theorem

Theorem 2 (The lift burden moves below the recorded lift package)**.** *Assume the fixed primitive continuation data of Definition 1 and, for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, lift-generating substrate precursor dynamics in the sense of Definition 3. Then:*

1. *the shell-generating substrate lift dynamics are produced unchanged by projected exact-shell transport from the deeper precursor dynamics;*
2. *because the induced lift package is exactly the one already used by the shell-restriction inevitability theorem, the primitive substrate shell-restriction package and the previously recorded downstream reservoir-facing consequences remain available unchanged;*
3. *the same benchmark one-metric package remains fixed: the operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged;*

4. *consequently the remaining honest burden on the active line is deeper again: derive or justify the lift-generating substrate precursor dynamics themselves from still more primitive substrate structure, or else prove a sharper inevitability theorem at that precursor level.*

Proof. Item (1) is Proposition 1. Item (2) is the routing consequence of item (1): the present note reproduces exactly the shell-generating lift package that the immediately preceding shell-restriction theorem already uses as its input, so that theorem and its downstream consequences apply unchanged. Item (3) is Theorem 1. Item (4) is the project-level consequence of the previous items. □

Corollary 1 (What remains open after this theorem)**.** *After Theorem 2, the remaining honest burden is no longer to justify the shell-generating substrate lift dynamics themselves. Those dynamics are now the canonical projected exact-shell output of a sharper precursor class. What remains open is why the lift-generating substrate precursor dynamics should hold, or whether an even sharper inevitability theorem can remove that still deeper precursor-level freedom while preserving the same benchmark one-metric package and claim boundary.*

Proof. Theorem 2 removes the shell-generating lift package as the current unexplained stopping point. The next honest burden therefore lies strictly below it. □

8. Conclusion

The positive result of this note is narrower than a completed first-principles derivation and stronger than another lift-level restatement. The shell-generating substrate lift package is no longer left on the active line as a merely recorded deeper assumption. Starting from lift-generating substrate precursor dynamics, the note derives the lifted restriction section from precursor lift-fiber minimization, derives the exact lifted shell as the projected precursor shell, derives the fiber-minimizing substrate lift, transports the same lifted shell and support charts, transports the same lifted support shell and lifted support zero core, transports the same zero-core / hidden-seed decomposition, and preserves the same hidden charge, exact zero-response realization, solvability statement, and coercive control.

Because that induced lift package is exactly the one already used by the immediately preceding shell-restriction theorem, the gain is more than another same-output origin relay. The deeper precursor theorem removes the lift layer itself as the current unexplained insertion while leaving the shell-restriction theorem and its downstream reservoir-facing consequences available unchanged.

The scope remains disciplined. The benchmark package is unchanged: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. This is therefore an origin-side theorem inside the active derivational program of Æther-flow Interpretation of Relativity, not a basis for stronger promotion language. The next honest burden is deeper again: derive or justify the lift-generating substrate precursor dynamics themselves from still more primitive substrate structure, or else prove a sharper inevitability theorem there. Until then, adoption and derivation should continue to be kept sharply separated.